

DDTA minimal BAE ontology candidate - R2

Status: STRONG CANDIDATE / NOT ACCEPTED / BA1 OPEN

Derived by: BA1-T1 + BA1-T2

Repository baseline reviewed: f05fbb7b253392e158a1062df2614b177c13d43e

Supersedes for active BA1 reasoning: BA1_MINIMAL_BAE_ONTOLOGY_CANDIDATE_R1.md

R1 remains immutable research history.

Candidate ontology

The current minimal candidate contains exactly two semantic identity families:

BAReferent

BAProposition

No domain-specific subtype split is accepted yet.

BAReferent

A **BAReferent** is an independently identifiable unit of methodology-neutral shared project meaning that must be reusable across one or more analytical propositions, projections or governed baselines.

A referent may denote structural or human/external meaning, information or resources, behavior or events, contracts, storage, boundaries, states or contexts, or other project meaning.

Those semantic kinds may be important classifications, but BA1-T2 does not promote them to dedicated first-class metaclasses.

When a behavior, event, milestone, state or other meaning must be independently reused or constrained, that meaning can itself receive **BAReferent** identity.

BAProposition

A **BAProposition** is an independently identifiable methodology-neutral analytical assertion about one or more **BAReferent** identities, with enough identity to support origin/provenance, diagnosis, reuse and change disposition.

A proposition may express project meaning such as:

- relation/dependency;
- behavior/action/transition semantics;
- responsibility/externality;
- correlation;
- state/condition/result semantics;
- applicability;
- constraint/property;
- lifecycle/failure/concurrency semantics.

The exact proposition structure, predicate/action vocabulary, role vocabulary and cardinalities remain BA2 questions.

Refined family boundary

A **BAProposition** is not itself the project meaning that it describes.

If project meaning must be independently reused or qualified, that meaning receives **BAReferent** identity and propositions state its semantics.

Therefore proposition-as-project-target is **not required by the minimal candidate**.

Proposition identity still exists so that BA3 can attach origin/provenance, diagnostic state and change disposition to the assertion itself without confusing those concerns with referent identity.

Minimal picture

Base Analysis for one governed baseline

BAReferent *

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| referenced by

|

BAProposition *

project-semantic reuse/qualification -> BAReferent

assertion provenance/diagnostic state -> BAProposition metadata in BA3

BAE remains an umbrella term, not an additional required metaclass.

Split-or-collapse result

BA1-T2 did not force dedicated first-class roots for:

- Participant / Actor;
- Component / Capability;
- Information / Resource;
- Behavior / Event / Transition;
- Interface / Connection / Channel;
- Store / Persistence;
- Contract;
- Boundary / Externality;
- State / Mode / Context;
- Dependency / Allocation;
- Property / Constraint;
- DataFlow / Interaction.

A concrete instance of any of those meanings may still be a **BAReferent** when independent identity is needed.

The opposite collapse into one undifferentiated semantic family is rejected because referent identity and assertion identity/origin/evolution are independently required.

A single programming class, table or serialization container with a mandatory semantic discriminator would be only a representation choice and would not collapse the two semantic families.

Document-layer boundary

Document types remain source-layer governed concepts and are not copied into Base Analysis as BAE types:

- MacroRequirement;
- Decision;
- Requirement;
- FunctionalRequirement;
- SpecializedRequirement;
- SecurityRequirement;
- NormativeClause.

Governed documentation remains project authority.

Deferred rather than promoted

The following remain required later responsibilities but are not domain BAE metaclasses in this candidate:

- source locator and provenance materialization -> BA3;
- grounded/derived/diagnostic materialization -> BA3;
- identity/equivalence/lifecycle mechanics -> BA3;
- views/projections -> BA4;
- controlled vocabulary and lexical assistance -> BA5;
- Base Analysis regression/closure -> BA6;
- AnalysisRecord, Finding and method-specific outputs -> later analysis envelope.

First-class split criterion

A new first-class BAE type/family is justified only if concrete evidence shows both:

1. independent semantic identity is required across propositions/projections/change; and
2. reusable subtype-specific invariants cannot be represented honestly as classifications, roles, values or propositions over the existing families.

Recurring usefulness, diagram convenience, implementation classes and method-specific categories are insufficient.

Falsification rule

The two-family candidate fails if a concrete governed corpus or method-neutral consumer requires a deferred semantic kind to have independent subtype-specific identity/invariants that cannot be represented honestly with `BAResult` + `BAProposition`, or if proposition identity can be removed without losing claim-level provenance/change/diagnostic semantics.

Next step

BA1-T3 - minimal BAE ontology closure review.

BA1 remains open. BA2 remains not started.